Determine the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit of Fig. 13.13. **Practice Problem 13.2**

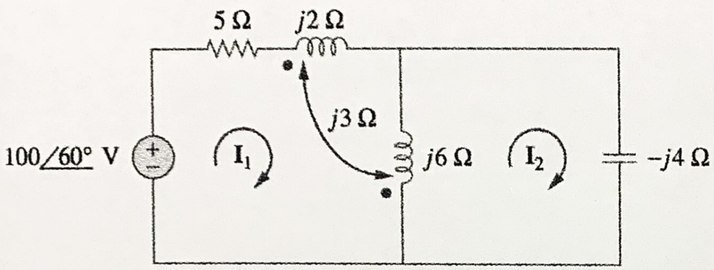
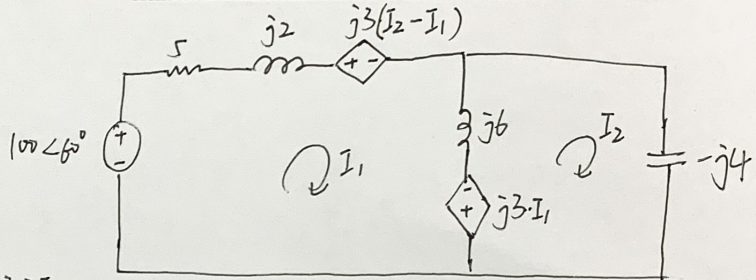


Figure 13.13
For Practice Prob. 13.2.

Answer: $\mathbf{I}_1 = 17.889\angle 86.57^\circ$ A, $\mathbf{I}_2 = 26.83\angle 86.57^\circ$ A.

互感的概念~稍复杂情况
Focus on 流过线圈的电流



$$-100\angle 60^\circ + (5+8j)I_1 - 6j\cdot I_2 + 3j(I_2-I_1) - 3jI_1 = 0$$

$$3jI_1 + (I_2-I_1)\cdot 6j - 4jI_2 = 0$$

$$\Rightarrow \begin{bmatrix} 5+2j & -3j \\ -3j & 2j \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 100\angle 60^\circ \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 17.889\angle 86.565^\circ \\ 26.833\angle 86.565^\circ \end{bmatrix} A$$

Practice Problem 13.3

For the circuit in Fig. 13.18, determine the coupling coefficient and the energy stored in the coupled inductors at $t = 1.5$ s.

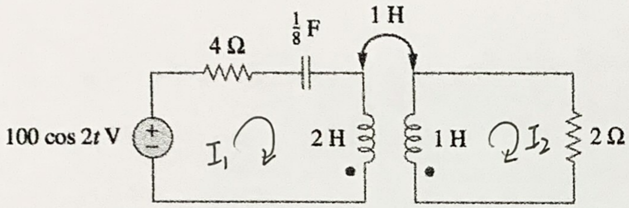


Figure 13.18
For Practice Prob. 13.3.

Answer: 0.7071, 246.2 J.

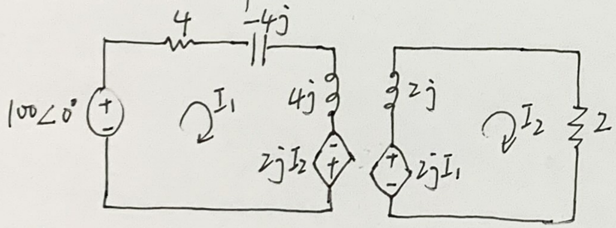
磁耦合系数和储能

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

$$\boxed{w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2}$$

① $K = \dfrac{M}{\sqrt{L_1 L_2}} = \dfrac{1}{\sqrt{2}} = 0.707$

② 在频域分析，$\omega = 2$，$\frac{1}{8}F \Rightarrow -4j$；$2H \to 4j$；$1H \to 2j$



$$-100\angle 0° + 4I_1 - 2jI_2 = 0$$

$$-2jI_1 + (2+2j)I_2 = 0$$

$$\Rightarrow \begin{bmatrix} 4 & -2j \\ -2j & 2+2j \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 100\angle 0° \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 19.612\angle 11.31° \\ 13.868\angle 56.31° \end{bmatrix}$$

$i_1(t) = 19.612 \cdot \cos(2t + 11.31°)$

$i_2(t) = 13.868 \cdot \cos(2t + 56.31°)$

当 $t = 1.5s$ 时

$W = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 - M i_1 i_2$

$= 245.099$ J

Practice Problem 13.4

Find the input impedance of the circuit in Fig. 13.25 and the current from the voltage source.

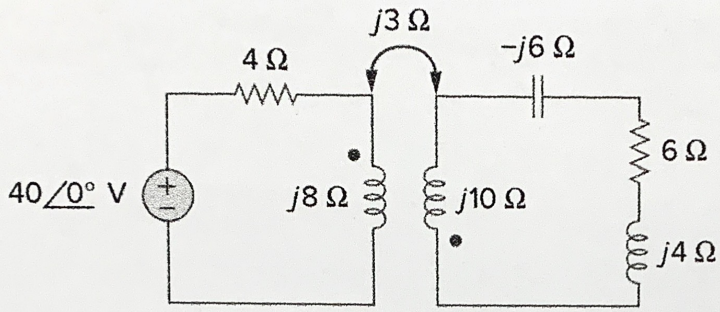


Figure 13.25
For Practice Prob. 13.4.

阻抗折算

$$\boxed{\mathbf{Z}_R = \frac{\omega^2 M^2}{R_2 + j\omega L_2 + \mathbf{Z}_L}}$$

Answer: $8.58\underline{/58.05^\circ}\ \Omega$, $4.662\underline{/-58.05^\circ}$ A.

次级回路总阻抗：$6 + 8j$

ωM：$3j$

$$\Rightarrow Z_{in} = 4 + 8j + \frac{3^2}{6+8j} = 8.58\angle 58.051^\circ$$

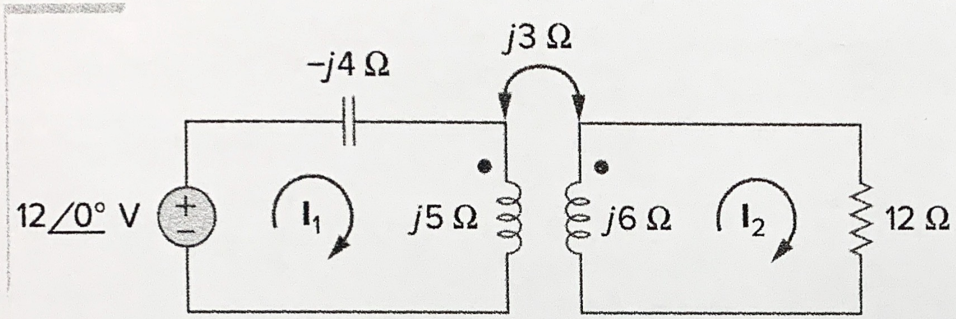
$$I = \frac{40\angle 0^\circ}{Z_{in}} = 4.662\angle -58.051^\circ$$

Practice Problem 13.6

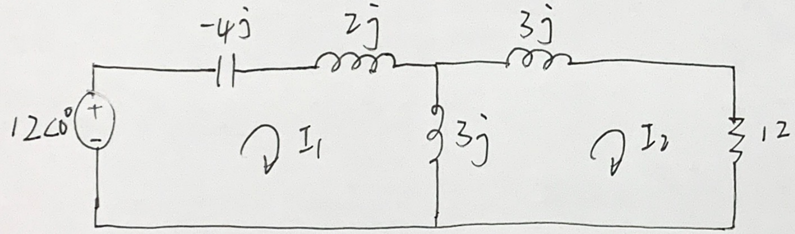
Solve the problem in Example 13.1 (see Fig. 13.9) using the T-equivalent model for the magnetically coupled coils.

Answer: $13\angle -49.4^\circ$ A, $2.91\angle 14.04^\circ$ A.

Calculate the phasor currents $\mathbf{I}_1$ and $\mathbf{I}_2$ in the circuit



去耦合等效法



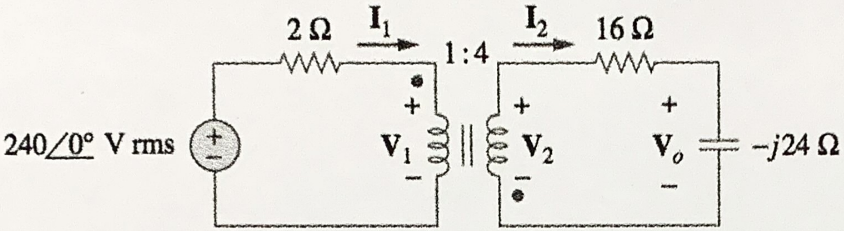
$$-12\angle 0^\circ + 1j\cdot I_1 - 3j I_2 = 0$$

$$(12+6j) I_2 - 3j I_1 = 0$$

$$\begin{bmatrix} 1j & -3j \\ -3j & 12+6j \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 12\angle 0^\circ \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 13.016\angle -49.399^\circ \\ 2.91\angle 14.036^\circ \end{bmatrix} A$$

Practice Problem 13.8

In the ideal transformer circuit of Fig. 13.38, find $\mathbf{V}_o$ and the complex power supplied by the source.



理想变压器的折算法简化

Figure 13.38
For Practice Prob. 13.8.

Answer: $429.4\angle 116.57^\circ$ V, $17.174\angle -26.57^\circ$ kVA.

将 16−24j 折算到初级 $Z_{in} = \dfrac{16-24j}{16} = 1 - 1.5j$

$\therefore I_1 = \dfrac{240}{3-1.5j} = 71.554\angle 26.565^\circ$

$I_2 = -\frac{1}{4} I_1$

$V_0 = I_2 \cdot (-24j) = 429.325\angle 116.565^\circ$

$P = V \cdot I^* = 240 \cdot 71.554\angle -26.565 = 1.717\angle -26.565 \times 10^4$ VA

Practice Problem 13.9

Find $\mathbf{V}_o$ in the circuit of Fig. 13.40.

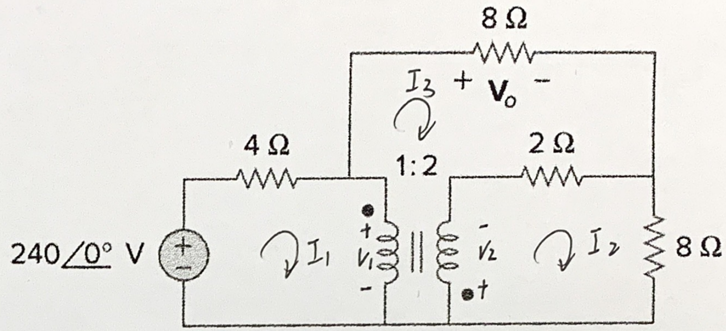


Figure 13.40
For Practice Prob. 13.9.

Answer: 96 V.

折算法简化不能应用时，
采用传统电路分析法

$$\frac{V_1}{V_2} = \frac{1}{2} \quad ; \quad \frac{I_1 - I_3}{I_2 - I_3} = -\frac{2}{1}$$

$$-240 + 4I_1 + V_1 = 0$$

$$V_2 + 10I_2 - 2I_3 = 0$$

$$-V_1 + 8I_3 + 2(I_3 - I_2) - V_2 = 0$$

$$2V_1 - V_2 = 0$$

$$I_1 - I_3 + 2(I_2 - I_3) = 0$$

$$\begin{bmatrix} 1 & 0 & 4 & 0 & 0 \\ 0 & 1 & 0 & 10 & -2 \\ -1 & -1 & 0 & -2 & 10 \\ 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -3 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 240 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ V_2 \\ I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 44.308 \\ 88.615 \\ 48.923 \\ -6.462 \\ 12 \end{bmatrix}$$

$$\therefore V_o = I_3 \cdot 8 = 96\ V$$